

Improved ICP Matching Algorithm Based on Laser Radar and IMU

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Abstract: In this paper, we proposed the improved iterative closest point algorithm based on laser radar and IMU. The system composition is described. The initial pose estimates of two scanning point clouds are obtained by IMU, and extraction of line feature of two 2-D point sets is put forward to accomplish the matching of fast changing scenario. The experiment is carried out, and the results show that the improved ICP algorithm could accelerate the iteration of ICP and improve the performance of point clouds.

Keywords: Laser radar; SLAM; Iterative closest points; Pose estimate

1 Introduction

Indoor mobile robot (IMR), as a kind of automatic intelligent equipment, has attracted more and more attention. With the increase of human cost and the improvement of automation and intelligence in the production process, demand for IMR will be more urgent for enterprise. Practically, laser navigation, as a crucial method to realize simultaneous localization and mapping (SLAM), based on laser radar has been widely studied and successfully applied in IMR [1-3]. The problem of SLAM is more closely related to environmental representation, map building, location technology, path planning and motion control.

The SLAM technology is mostly divided into two methods, which are the probability estimation method and the non probability estimation method. The former applies probability theory to the positioning process of mobile robot. The typical methods are ICP SLAM [2, 4], extended Kalman filter (EKF) SLAM [5, 6] and particle filter (PF) SLAM [7]. The latter is Graph SLAM [8]. Presently, the time costs of EKF-SLAM, PF-SLAM and Graph-SLAM are enormous on account of the complex operation process. Especially, in [5, 6], the intelligent optimization methods are incorporated into the SLAM algorithm. It is difficult to realize these algorithms in practical. On contrary, the ICP-SLAM is realized only using the matching of two frames point clouds, which was verified in past years. However, the traditional ICP method needs to be improved because of the lack of initial pose estimate and effective matching.

In this paper, the improved iterative closest points based on laser radar and IMU is proposed. In Section 2, the IMR system composition is described. Section 3

improved the ICP method by initial pose estimate and extraction of line feature. In Section 4, the real experiment is carried out to verify the improved ICP method. Finally, some conclusions are achieved.

2 The system composition

The block diagram of hardware composition for laser navigation IMR system is shown in Figure 1. The sensor system is divided into internal and external sensors. The external sensor is two-dimensional laser radar, which could perceive environment and obtain the distance and angle information. The external sensors consist of two encoders, IMU and an electronic compass. Two encoders are mounted on the driving wheel of IMR to estimate the pose. Although it is inevitable for drift of IMU integral measurement and effect of magnetic field for electronic compass, the fusion algorithm could be adapted to make up this defect and improve the accuracy of position measurement.

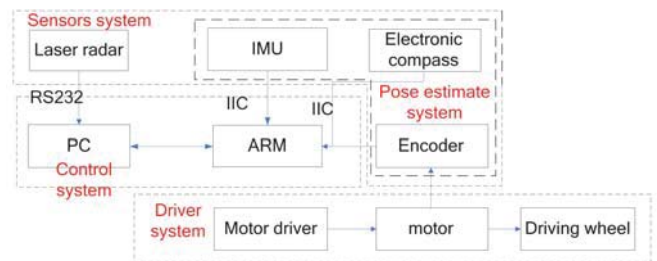


Figure 1 Block diagram of hardware composition for laser navigation IMR system

3 The improved ICP Method

The key problem in laser navigation SLAM is to solve the problem of frame matching. The most commonly used method is ICP matching. We give two 2-D point sets $\{p_i\}: i=1,2,\dots,N$, (here, p_i and p'_i are considered as 2×1 column matrices)

$$p'_i = Rp_i + T + N_i \quad (1)$$

where R is a 2×2 rotation matrix, T is a translation vector (2×1 column matrix), and N_i is a noise vector. We want to find R and T to minimize

$$\Sigma^2 = \sum_{i=1}^N \|Rp_i + T + N_i\|^2 \quad (2)$$

We use the singular value decomposition (SVD) to obtain the optimal estimate. It is significant to select the initial rotation matrix R and pose vector T [9]. Thus, we achieve the initial data through inertial measurement unit (IMU). The Improved ICP scan matching is shown in Figure 2. The whole scheme is divided into two parts, and the process is as follows:

1. The IMU and compass are used to acquire the initial pose estimate.
2. The numbers of 2-D point sets are compared to judge fast changing scenario.
3. If the scenario changes fast, the local straight line is extracted from the original laser radar data. The initial pose estimate and local straight line are applied to ICP matching to improve the accuracy of matching.
4. Contrarily, the only initial pose estimate is adopted to improve the ICP matching.

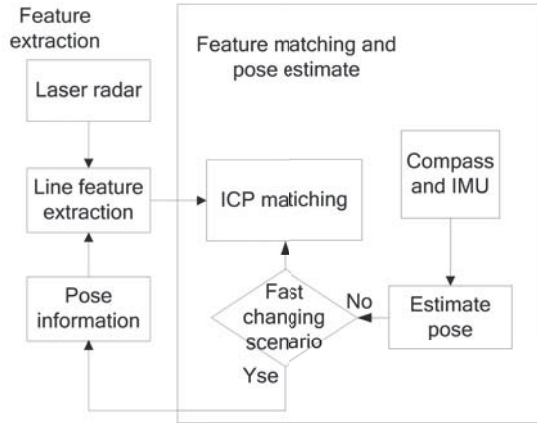


Figure 2 Diagram of Improved ICP scan matching image.

4 Experiments and analysis

The experiment prototype is shown as Figure 3. The industrial personal computer and battery mount on the platform. The bracket is utilized to connect to laser radar, whose type is rplidar A2 of SLAMTEC Corporation. The experiment data collected by Linux ubuntu 14.04 is employed to verify effectiveness of proposed method.



Figure 3 The indoor navigation prototype

ICP matching analysis is carried out between two frames. In the first case, the scenario changes little. Thus, the displacement is relatively small. As shown in Figure 4, the initial parameters estimated using the pose calculation of IMU are:

$$R = \begin{bmatrix} 0.9998 & 0.0197 \\ -0.0197 & 0.9998 \end{bmatrix} \quad (3)$$

$$T = [-0.1942 \quad -0.8231]^T$$

It can be seen from the graph that the initial pose estimation could be effectively realized adding the initial matrix. Two point clouds almost overlap. Therefore, the convergence speed of the ICP can be faster. The experiments after iteration of the ICP algorithm are shown in Figure 5. It can be seen from the diagram that the two graphs are almost completely overlapped. The rotation matrix and the translation vector after calculation are as follows:

$$R_e = \begin{bmatrix} 0.9995 & -0.0301 \\ 0.0301 & 0.9995 \end{bmatrix} \quad (4)$$

$$T_e = [-6.52e-4 \quad 0.0854]^T$$

The rotation matrix is similar. Nevertheless, the plane shift vector is different by comparing the initial parameters and the parameters after the iteration. Because the IMU sensor obtains the rotation matrix through the angular velocity integral, and the error is relatively small. However, The cumulative error is relatively large on account of acceleration of two times integral. From Figure 6, it can be seen that the selection of the initial estimation of the position and attitude parameters can effectively reduce the error of ICP iterative function. Compared with traditional ICP, the improved ICP has converged after third iterations. Whereas, the iteration of position estimation cannot converge until the iteration reaches eighth times. Meanwhile, we can see that because of the same conditions of convergence, the objective function value after the final iteration is similar.

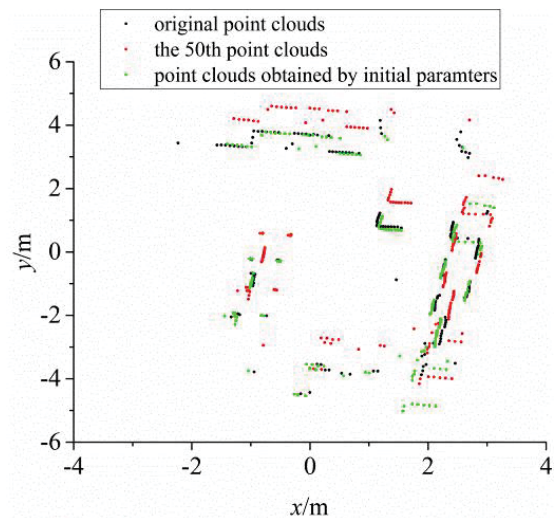


Figure 4 The initial point clouds

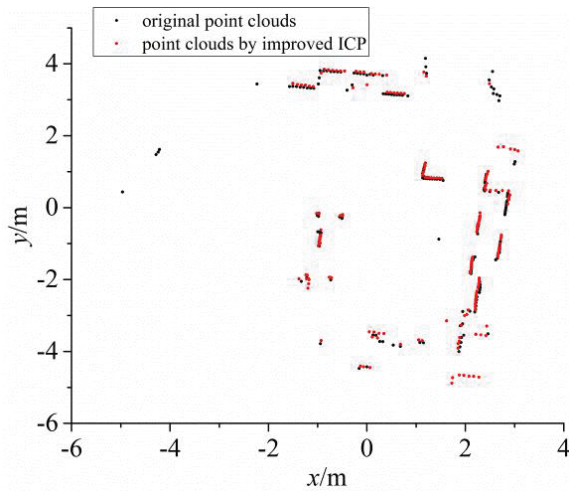


Figure 5 The comparisons of ICP algorithm

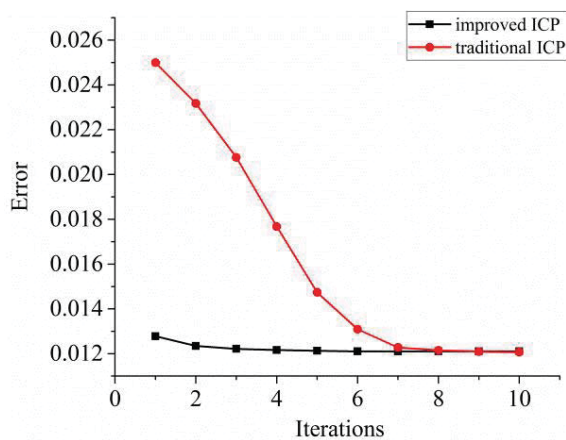


Figure 6 Errors with different ICP

In the second case, the scenario is shown in Figure 7. It can be seen that half of point clouds don't overlap. The directly used ICP method will cause large error. The results are shown in Figure 8. The matching result is not ideal. Thus, we extract the line feature. By these features, we could acquire optimal pose estimate better. In this figure, only the left part could be used to extract line feature to achieve the estimate pose. The results could be seen as in figure 9. Two 2-D point sets could be matched according to the only partly overlapping point clouds.

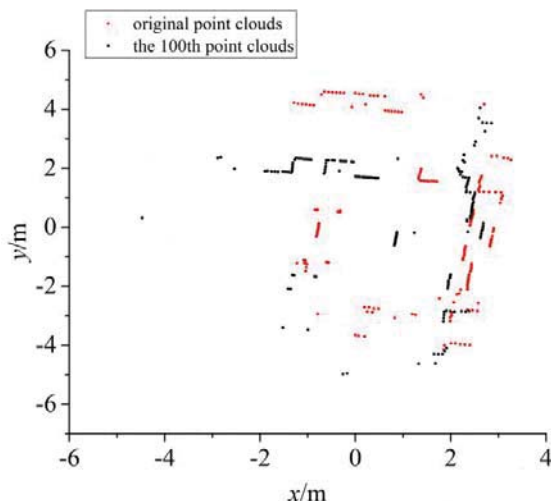


Figure 7 Errors with different ICP

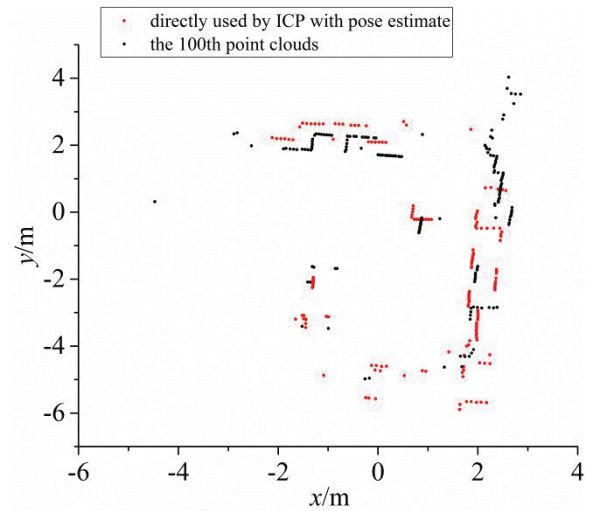


Figure 8 the results without line feature extraction

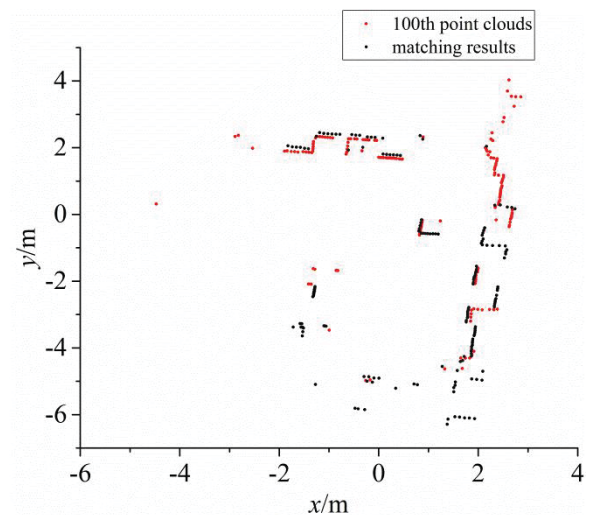


Figure 9 the results with line feature extraction

5 Conclusions

The improved ICP algorithm based on laser radar and IMU is proposed. Some conclusions can be achieved: (1) compared with traditional ICP, the improved ICP has converged after third iterations. The convergence speed is accelerated. (2) The accuracy of ICP matching is assured by fusion of IMU data and line feature of laser radar data.

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